Safety-Gated, Age-Adaptive LoRA Fine-Tuning for a Virtual Pediatric Hospital Guide

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Abstract

Pediatric patients need age-appropriate, institutionally grounded information that general-purpose AI systems are not designed to provide, and any system accepting free-text input from children must defend against inputs it was never designed to handle. We present *small-bear*, an age-targeted conversational guide deployed within a digital recreation of Victoria Hospital (London, Ontario), paired with our *guard-bear*, an upstream input-safety classifier – together a complete pipeline running entirely on consumer hardware (an Apple Mac Mini M4, 16 GB) with no cloud dependency. The first finding concerns safety: the unmodified base classifier we build on, `Prompt-Guard-86M` [1], performs *worse than random* (ROC-AUC 0.448) on pediatric-context queries, saturating by flagging nearly everything as unsafe. Fine-tuning *guard-bear* on 4,053 pediatric-specific labeled examples (deduplicated, verified disjoint train/val/test) resolves this to near-perfect discrimination, confirmed across a 5-seed retraining campaign (ROC-AUC $\mathbf{0.9992{\pm}0.0005}$, recall $\mathbf{0.988{\pm}0.004}$) – evidence that safety mechanisms validated for a developer-facing threat model do not transfer to a different population without dedicated evaluation. The second finding concerns *small-bear*: we fine-tune Llama 3.2 Instruct (1B, 3B) with LoRA at two configurations, Standard ($\boldsymbol{r = 8}$, 16 layers) and Fast ($\boldsymbol{r = 4}$, 8 layers), on 1123 LIMA-inspired [2] synthetic examples, substantially outperforming zero-shot base models on readability for the 5–11 role (56–66% FK $\leq$ 7.0 pass rate vs. 12%; FK $\leq$ 7.0 is not evaluated for 12–18) and inter-role lexical separation (0.92–0.96 vs. 0.64–0.67). A cheaper prompted-but-not-fine-tuned baseline matches or exceeds these metrics (92–98%, 0.970–1.000) but at 1.2–2.2× the tokens/response, failing the 1.0-second latency target it costs nothing to attempt; one explicit length clause closes that gap on 1B entirely (beating every adapter) but not 3B, and costs inter-role separation on 1B (0.870 vs. 0.900–0.960) – fine-tuning's most defensible remaining edge is reaching the target without a prompt hand-tuned to the evaluation's own thresholds, with Fast-1B the only such configuration meeting it on average for the 5–11 role. During development, an unexpected **configuration-ordering crossover** emerged (Standard better on 3B, Fast better on 1B); a dedicated 110-run multi-seed campaign confirmed it is statistically real on both sides, but not real in the same way: the 1B-side gap is a length-independent register effect, while the 3B-side gap, once length is controlled for, is only marginal ($\boldsymbol{p = 0.055}$) and substantially attributable to response length rather than register. Together, this pipeline's components each run practically on consumer hardware (combined, end-to-end latency remains unmeasured), and its safety layer, not response quality, posed the larger practical risk when unvalidated.

Keywords: parameter-efficient fine-tuning, low-rank adaptation, pediatric clinical NLP, virtual reality, on-device inference, small language models, LLM input safety classification

1 Introduction

Large language models (LLMs) enable domain-specific conversational systems, but resource-constrained clinical deployment raises constraints rarely addressed together: limited memory favors smaller models, domain adaptation requires fine-tuning, real-time interaction imposes strict latency bounds, and free-text input from a vulnerable population must be defended against inputs the system was never designed to handle.

This paper presents *small-bear*, an age-targeted conversational guide for children aged 5–11 and adolescents aged 12–18 embedded in a VR recreation of Victoria Hospital (London, Ontario), together with *guard-bear*, an upstream safety classifier – both named after LHSC's bear mascot. All training and inference runs on an Apple Mac Mini M4 (16 GB): consumer hardware, no cloud infrastructure.

A conversational system accepting open-ended text from children needs a defense layer distinct from *small-bear* itself. General-purpose prompt-injection classifiers such as `Prompt-Guard-86M` [1] target a developer-facing threat model that does not describe a pediatric system's actual inputs: short, first-person legitimate queries on one side, gibberish and adult content on the other. Our *guard-bear* closes this gap, finding, as our central safety result, that the unmodified base classifier performs worse than random here – a failure fine-tuning fully resolves.

For *small-bear*, we fine-tune Llama 3.2 Instruct models [3] at 1B and 3B scales using LoRA [4], alongside QLoRA [5]. Training data follows the LIMA hypothesis [2]: $\sim$560 examples per adapter (1123 total) from Victoria Hospital's public materials. We compare Standard ($r = 8$, 16 layers) and Fast ($r = 4$, 8 layers) adapters across both sizes and age groups. Pediatric deployment demands developmental vocabulary, supportive register, and hard safety constraints distinct from adult-facing clinical LLMs; we are not aware of prior work fine-tuning age-stratified LoRA adapters for this setting (Section 2).

During development, an unexpected **configuration-ordering crossover** emerged (Standard better on 3B, Fast on 1B) – opposite existing rank-selection guidance at the 7B–70B scale – validated with a 110-run multi-seed campaign (Section 3.3) after an initial single-seed read proved non-reproducible.

The contributions of this paper are:

- *guard-bear*: a fine-tuned input-safety classifier showing its unmodified general-purpose base model performs *worse than random* on this population – resolved by fine-tuning to near-perfect, 5-seed-validated discrimination – evidence that safety classifiers cannot be assumed to transfer to a new population without dedicated evaluation.
- A complete age-adaptive pipeline, gated by an upstream safety classifier, trained end-to-end on consumer hardware; a prompted, non-fine-tuned baseline – including a length-capped variant that matches the adapters on 1B – isolates fine-tuning's real contribution as not depending on a threshold-aware prompt, plus a style-separation gap (Section 3.1).
- An unexpected LoRA configuration-ordering crossover between Llama 3.2 1B and 3B, confirmed statistically real on both sides by a 110-run multi-seed campaign after an initial single-seed observation proved non-reproducible – though length-adjustment shows the 3B side is substantially a length effect, unlike 1B (Section 3.3).
- A rank sweep and layer sweep investigating the crossover's mechanism (Section 3.4), and cross-architecture evidence (Qwen 2, SmolLM2, Qwen 2.5) that the Fast-vs-Standard directional pattern generalizes beyond Llama even where the rank-regime mechanism does not cleanly resolve.

2 Related Work

General-purpose prompt-injection classifiers such as `Prompt-Guard-86M` [1] target developer-facing override patterns; the closest child-safety effort we found, *KidRails* [6] (an open-source project, not a peer-reviewed study), fully fine-tunes an 8B response model without a separate classifier or baseline comparison. *guard-bear* instead evaluates a dedicated input classifier against its base model on a labeled pediatric threat taxonomy – we are not aware of a peer-reviewed precedent for this population.

Prompted, not fine-tuned, LLMs have been evaluated as pediatric communication aids: Rao et al. [7] found off-the-shelf models can produce age-appropriate surgical explanations but called for further fine-tuning; a Norwegian evaluation similarly found generated child-directed language skewed too advanced [8]. *KidLM* [9] adapts masked-language-model pretraining to a children's corpus but targets general modeling, not clinical generation. VR-based interventions reduce procedural anxiety [10] but are passive, scripted content; no prior work, to our knowledge, fine-tunes age-stratified LoRA adapters for pediatric hospital communication.

LoRA [4] and QLoRA [5] are standard for resource-constrained fine-tuning, but rank-selection guidance – including Biderman et al.'s [11] empirical study, the most thorough to date – is validated only at the 7B–70B scale; rank-ordering below 7B is uncharacterized by prior work. Our rank and layer sweeps (Section 3.4) address this gap.

3 Results

3.1 small-bear: Perplexity, Readability, Latency, and Style Separation

Table 1 reports a single seed-42 run of the four fine-tuned configurations and both base models; Section 3.3's properly-powered campaign is the paper's actual evidentiary basis for the crossover and is authoritative wherever the two differ. All four fine-tuned adapters substantially outperform zero-shot base models on FK ≤ 7.0 for the 5–11 group (Table 1: 56–66% vs. 12%) and inter-role style separation, not tabulated but computed identically to Table 5's Style-Sep. column (Std-1B/3B, Fast-1B/3B: 0.950/0.930/0.960/0.920 vs. Base-1B/3B: 0.670/0.640) – though this classifier does not discriminate Standard from Fast, consistent with, not independent confirmation of, the readability pattern below. **But zero-shot is not the relevant alternative**: the same base model given one age-targeted system-prompt sentence[1], no fine-tuning – single seed-42 run, one untuned prompt per role, not the multi-seed treatment applied to the crossover – reaches 92–98% FK$\leq$7.0 (Table 1, Prompted-1B/3B rows) and 0.970–1.000 style separation (same metric, not tabulated), matching or beating every adapter. It does not control length: 145–250 tokens vs. the adapters' 87–163 (same-role comparisons range 1.2–2.2$\times$), pushing latency to 2.0–10.3s and leaving only 0–8% of turns under 1.0s vs. Fast-1B's 62%. Is that just a badly-engineered prompt? Adding one clause – "Keep your answer under 60 words" – to that prompt (Prompted-Brief, not tabulated) closes the 1B gap entirely (100% FK$\leq$7.0, 0.64s avg, 100% under 1.0s, beating every adapter) at a style-separation cost (0.870 vs. 0.900–0.960, nearer Base's 0.670–0.640). On 3B the clause still fails the target (1.48s avg, 0% under 1.0s, down from 5.03–10.29s but still 4–8$\times$ over) despite comparable FK-pass (94%) and separation (0.940). This is the strongest evidence against fine-tuning's necessity here, reported because it is real – but two things limit its weight. It is single-seed ($n = 50$), unlike this paper's other headline claims, so whether it replicates is untested (Section 3.3). And the clause required us to already know the FK/latency thresholds – a best-case, threshold-aware prompt, not evidence prompting is robust to phrasings a deploying engineer would pick blind (Limitations). We do not think the style-separation gap alone (0.870 vs. 0.900–0.960) settles this – that same classifier is shown above not to discriminate Standard from Fast, so its sensitivity here is itself unresolved. **What survives, pending replication, is narrower than originally framed**: this paper cannot claim fine-tuning beats a threshold-aware prompt on 1B readability or latency, only that it avoids depending on that foreknowledge. Among configurations not already tuned to known thresholds, **Fast-1B remains the only fine-tuned configuration meeting the 1.0s VR latency target on average** (0.97s; 62% of turns individually, Table 1); all 3B configurations exceed 2.2s. Post-hoc, single-seed perplexity does not track FK-based readability cleanly – Fast is lower than Standard at *both* sizes (1B: 30.18 vs. 30.46; 3B: 22.91 vs. 24.88) – reported only as a construct-validity flag, not a confirmed finding absent multi-seed treatment.

On this seed, Fast-1B beats Standard-1B on FK pass rate (62% vs. 56%), while Standard-3B and Fast-3B tie (66% vs. 66%); Section 3.3's campaign shows this single-run picture undersells the true, smaller 3B-side gap – the qualitative direction holds, but treat the margins here as illustrative, not as the paper's statistical evidence.

3.2 guard-bear: Safety Classification Results

The clearest evidence that general-purpose safety tooling cannot be assumed to transfer to a new population comes from comparing *guard-bear* against unmodified `Prompt-Guard-86M` (threshold 0.50) on the 608-example held-out test set (Table 2), verified zero text overlap with training (Section 4.3). **The base model is worse than random**: ROC-AUC 0.448 is below chance, since its imperative-override feature space does not describe short, first-person pediatric queries or gibberish/out-of-scope/adult content. Its apparent 0.9385 recall is a saturation artifact, flagging 292 of 299 safe examples as unsafe too; the score is $1 - P(\text{benign})$ from its native softmax, ruling out a

[1] Age 5–11: "You are Dr. Beary Good, a warm and friendly guide at a children's hospital. You are talking to a child between the ages of 5 and 11. Use simple words, short sentences, and a warm, reassuring tone." Age 12–18 substitutes "a teenager between the ages of 12 and 18. Use clear, respectful language appropriate for a teen audience." Untuned, single prompt per role (Limitations).

Table 1 Readability (50 age-matched held-out prompts/group) and latency/throughput (sequential, Mac Mini M4), single run (seed 42), current dataset vintage. FK≤7.0 is calibrated to the younger band only. Prompted-{1B,3B} is the same zero-shot base model with a simple age-targeted system prompt (no fine-tuning) – the cheap alternative baseline discussed in Section 3.1.

Config.	Age	FK	SMOG	G.Fog	C-L	TTR	FK≤7.0	Avg Lat.(s)	<1.0s
Standard-1B	5–11	7.1	9.2	8.7	7.1	0.756	56%	1.12	26%
Standard-1B	12–18	10.3	12.7	13.0	10.7	0.734	—	1.36	2%
Standard-3B	5–11	6.6	8.9	8.1	7.1	0.781	66%	2.22	0%
Standard-3B	12–18	10.4	12.8	12.9	10.6	0.733	—	3.08	0%
Fast-1B	5–11	6.5	9.0	8.0	6.8	0.752	62%	0.97	62%
Fast-1B	12–18	9.7	12.2	12.1	10.0	0.725	—	1.11	38%
Fast-3B	5–11	6.5	9.0	8.1	7.1	0.724	66%	2.70	0%
Fast-3B	12–18	10.0	12.5	12.5	10.4	0.663	—	3.63	0%
Base-1B	5–11	9.5	11.9	12.0	9.6	0.582	12%	1.90	16%
Base-1B	12–18	10.4	12.9	13.1	11.6	0.553	—	2.23	2%
Base-3B	5–11	9.8	12.4	12.5	10.2	0.609	12%	4.54	8%
Base-3B	12–18	11.7	13.6	14.7	11.4	0.551	—	5.40	2%
Prompted-1B	5–11	4.5	7.1	5.8	5.2	0.612	98%	1.97	8%
Prompted-1B	12–18	7.4	10.0	9.3	7.7	0.569	—	2.22	2%
Prompted-3B	5–11	4.7	7.5	6.2	5.6	0.661	92%	5.03	0%
Prompted-3B	12–18	8.4	11.3	10.6	8.8	0.581	—	10.29	0%

label-flip artifact. **Fine-tuning resolves the discrimination failure consistently**: a 5-seed campaign (Table 2, rightmost column) gives ROC-AUC 0.9992 ± 0.0005, every seed meeting all four deployment targets (recall/precision/F1/accuracy ≥ 0.97/0.85/0.90/0.92) – seed 42 is representative, not cherry-picked. A cheap TF-IDF+logistic-regression baseline, trained on the identical 2,836-row set guard-bear's gradients see, is not trivial (0.9911 AUC) but is measurably below the transformer, with 4–5× as many combined errors (32 vs. 6 at seed 42, ∼7.8/run over 5 seeds): fine-tuning adds real discriminative value, not just an easy bar (no public-sourced *safe* class was tested, though; Limitations). Largest subcategory gain over base: `gibberish_malformed` (38% → 92%). A source-provenance check (safe all synthetic; unsafe 59% synthetic, 41% public corpora) finds 98.3% recall on synthetic-source unsafe vs. 100% pulled-source – against provenance shortcut learning, though paraphrase leakage remains unchecked. One caveat: the *threshold* varies widely by seed (0.94 to 0.04) though performance does not – always re-tune after retraining.

Table 2 guard-bear vs. base model and a simple lexical baseline, held-out test set (608 examples: 299 safe, 309 unsafe; deduplicated, verified zero text overlap with train/val). TF-IDF+LR: logistic regression on TF-IDF features, trained on the same 2,836-row set guard-bear's gradients actually see (not the full train+val pool) and the same test set. Rightmost column: 5-seed campaign (42, 123, 456, 789, 2024), mean ± SD.

Metric	Base (t=0.50)	TF-IDF+LR	seed 42	guard-bear (5-seed)
Unsafe recall	0.9385	0.9644	0.9903	**0.9883±0.0043**
Unsafe precision	0.4983	0.9342	0.9903	**0.9864±0.0062**
Unsafe F1	0.6510	0.9490	0.9903	**0.9874±0.0049**
Accuracy	0.4885	0.9474	0.9901	**0.9872±0.0050**
ROC-AUC	0.4477	0.9911	0.9998	**0.9992±0.0005**

Confusion	Base	TF-IDF+LR	seed 42	5-seed sum (n=3040)
TN	7	278	296	1474
FP	292	21	3	21
FN	19	11	3	18
TP	290	298	306	1527

Table 1's latency figures and *guard-bear*'s footprint together show a complete pipeline practical on one consumer machine; Fast-1B alone clears the 1.0s VR target, as do several cross-architecture configurations (Table 5). End-to-end latency with *guard-bear*'s pass included has not yet been measured (Limitations).

3.3 Statistical Validation of the Crossover: A Multi-Seed Campaign

The single-run pattern above (Section 3.1) was an unexpected finding uncovered during development, not a designed comparison, so it needed the same reproducibility scrutiny as any surprising result before being treated as real. The ablation and the rank/layer sweeps in Section 3.4 are single- or two-seed comparisons; Section 4.3 documents why that scrutiny was necessary – a 17-point swing on a nominally identical re-run, traced primarily to a dataset revision plus a smaller, unexplained non-determinism effect.

Given this, we resolved the crossover question directly with a dedicated campaign: retraining Standard and Fast at their exact defined configurations ($r = 8$/`num_layers=16` and $r = 4$/`num_layers=8`) across many seeds, one consistent (current) dataset vintage throughout, role `age_5_11`, FK≤7.0 pass rate – $n = 25$/config at 1B (pre-committed, no interim look); at 3B, an initial $n = 15$/config batch left the comparison marginal ($t = 2.01$, $p = 0.054$), so we collected 15 further seeds/config as a single, capped top-up (not an open-ended stopping rule) and combine both stages below via an equal-weight inverse-normal combination test whose weights were fixed before analyzing the top-up batch – the valid correction when, as here, the extension decision itself was informed by the interim look rather than blinded to it.

Table 3 Seed-campaign FK≤7.0 pass rate (mean ± SD across independent training runs).

Config.	n	Mean	SD
Fast-1B	25	63.4%	6.3
Standard-1B	25	52.1%	6.3
Fast-3B	30	52.2%	7.0
Standard-3B	30	58.3%	6.9

On 1B, Fast beats Standard ($t = 6.40$, $p = 6.0 \times 10^{-8}$). On 3B, the naive pooled test gives $t = 3.42$, $p = 0.0011$, but the interim look makes this anti-conservative; combining that stage with the independent 15-seed replication collected afterward ($t = 2.75$, $p = 0.010$) via the combination test gives the valid $p = 0.0015$ – **the crossover survives the statistically correct test on both sides.** This campaign covers FK≤7.0 pass rate only, for `age_5_11` only; the other four readability metrics, the `age_12_18` role, latency, and the classifier are single-run, as in Table 1, though now on the same consistent dataset vintage as this campaign.

Because Standard and Fast differ systematically in response length (Table 1), we tested whether the crossover survives controlling for length, across all 5,500 individual campaign responses. Fast is shorter than Standard at 1B (72 vs. 78 words, $p = 4 \times 10^{-16}$) but longer at 3B (97 vs. 76 words, $p = 9 \times 10^{-50}$). Residualizing FK grade on a single word-count regression pooled across configs (an equal-slopes assumption we did not test) and comparing residuals by config, **the 1B crossover survives length-adjustment** (residual gap 0.35 FK grade levels, $t = -5.95$, $p = 3 \times 10^{-9}$), but **the 3B crossover is substantially a length effect** – the residual gap shrinks to 0.10

FK grade levels, only marginal ($t = -1.92$, $p = 0.055$). A joint logistic regression of pass/fail on config and word count agrees: the config coefficient stays large at 1B (0.35) but is near zero at 3B (0.007) once length is included. Standard's 3B advantage is better described as producing shorter, not more simply-worded, responses.

3.4 Mechanism Sweeps: Rank and Layer Depth

The Standard/Fast configurations differ in both rank and `num_layers`, so even with the crossover confirmed (Section 3.3), it is not attributable to either factor alone. Our working hypothesis is capacity regularization (higher rank permits more overfitting on this LIMA-scale dataset) [4, 11], though prior work does not characterize sub-7B behavior. The sweeps use two seeds – lower power than the campaign above; ±std is a two-point range, not a variance estimate. Both were retrained on the current vintage.

Fixing `num_layers=8` and sweeping $r \in \{2, 4, 8, 16\}$ across four architectures (Llama 1B/3B, Qwen 2 0.5B, SmolLM2 360M; two seeds; 32 runs; Table 4, left), **only SmolLM2 360M shows a clean, monotonic rank-regime pattern** (48% → 63%), consistent with depth driving a preference for higher rank; Llama – the architecture the crossover claim is actually about – and Qwen are flat or only weakly increasing within noise, so this sweep does not mechanistically explain the paper's own headline crossover for its primary model family. We treat SmolLM2's sensitivity as the sweep's one qualitatively consistent finding at $n = 2$/rank, and "rank is the operative variable" as suggestive, not resolved.

A companion layer sweep (fixed `rank=4`, `num_layers`$\in \{4, 8, 16\}$, Llama 1B/3B, two seeds, 12 runs; Table 4, right) replicates more consistently: **3B improves with depth** (57% → 63%) while **1B peaks at the fewest layers** (69% at layers=4, declining to 60% at layers=16), matching the original sweep's qualitative conclusion. Latency scales with layer count for both sizes.

Table 4 Rank sweep (left; num_layers=8 fixed) and layer sweep (right; rank=4 fixed) FK ≤ 7.0 pass rate, mean ± std across two seeds (n=2; read as a range between the two runs, not a variance estimate), current dataset vintage.

Model	r=2	r=4	r=8	r=16
Llama 1B	63±10	59±7	**64±6**	62±6
Llama 3B	54±11	58±3	57±1	**62±11**
Qwen 0.5B	70±3	**71±13**	71±4	60±11
SmolLM2	48±3	56±6	62±6	**63±4**

Model, layers	FK≤7	Lat.(s)	Tok/s
1B, 4	**69%**	0.85	**105.6**
1B, 8	61%	0.84	97.4
1B, 16	60%	1.06	84.0
3B, 4	57%	2.73	**45.0**
3B, 8	60%	2.67	43.2
3B, 16	**63%**	**2.24**	39.5

3.5 Cross-Architecture and Capacity-Ladder Validation

To test generalization beyond Llama, we evaluate Qwen 2 0.5B and SmolLM2 360M [12] and the Qwen 2.5 family (0.5B, 1.5B, 3B) for age 5–11 (Table 5), single-seed and exploratory. **SmolLM2 Standard dominates** (80% FK); **Qwen 2.5 shows Fast at or above Standard at every size**, suggesting no crossover for that family. The Style-Sep. column reflects the original, pre-revision vintage.

Table 5 Cross-architecture and Qwen 2.5 capacity-ladder results, age 5–11 (50 examples, seed 42, current dataset vintage; Style-Sep. is the inter-role style-separation classifier, 5-fold CV, chance = 0.50, †original pre-revision vintage, not yet re-validated).

Config.	FK≤7.0	Avg Lat.	<1.0s	Tok/s or Tok	Style-Sep.†
Qwen 2 Std.	68%	0.54 s	100%	83 tok	0.940 ± 0.049
Qwen 2 Fast	62%	0.51 s	96%	94 tok	0.960 ± 0.058
SmolLM2 Std.	80%	0.87 s	72%	82 tok	0.950 ± 0.032
SmolLM2 Fast	60%	1.30 s	56%	137 tok	0.920 ± 0.040
Qwen2.5-0.5B Fast	56%	0.45 s	100%	176.4	0.950 ± 0.032
Qwen2.5-0.5B Std.	56%	0.57 s	100%	152.1	0.940 ± 0.097
Qwen2.5-1.5B Fast	74%	1.07 s	58%	76.1	$\mathbf{0.980 \pm 0.024}$
Qwen2.5-1.5B Std.	66%	1.28 s	6%	67.8	0.890 ± 0.073
Qwen2.5-3B Fast	70%	1.73 s	4%	43.0	0.970 ± 0.040
Qwen2.5-3B Std.	50%	2.18 s	0%	39.9	0.930 ± 0.068

4 Methods

4.1 System Architecture

Raw text first passes through *guard-bear*, a binary input classifier; unsafe input is blocked before reaching *small-bear*. Safe input is routed by an age-gate, supplied by the calling VR application, to one of two LoRA adapters on a shared frozen 4-bit quantized base model, never fused into it. Emotional-support queries are in scope only as reassurance/redirection; clinical assessment and crisis intervention are out of scope. No system prompt is used, so the adapter weights encode register directly. Figure 1 shows the pipeline; the age_5_11/Fast/1B branch meets the 1.0s VR latency target (Section 5).

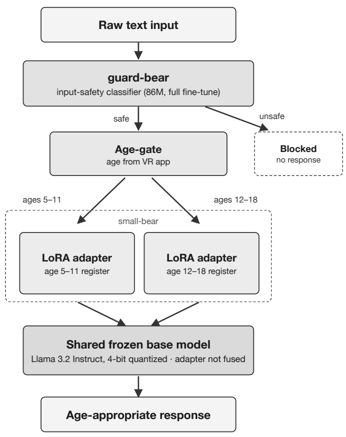

Fig. 1 System architecture. *guard-bear* (86M params, 42.05ms mean single-query latency, measured on the same Mac Mini M4) screens input before an age-gate routes safe queries to one of two LoRA adapters on a shared frozen 4-bit Llama 3.2 base; adapters are never fused. Rank/layer and base size are runtime choices behind each adapter box, not drawn separately – Table 1's four configurations are all instances of this diagram.

guard-bear classifies text as **safe** (a legitimate pediatric query, including benign out-of-scope curiosity) or **unsafe** (injection, jailbreak, adult/parental, gibberish, inappropriate, social engineering), a full parameter fine-tune of a small (86M) classifier, unlike *small-bear*. Design target: high recall, since a missed unsafe input is worse than a blocked legitimate one (≥ 0.97). Measured single-query latency (100 warmed-up forward passes, same Mac Mini M4, MPS backend): 42.05ms mean, comfortably under a 100ms target.

4.2 Datasets

small-bear's dataset was generated by Claude Opus 4.6 in extended thinking mode from the Children's Hospital Patient and Family Guide [13], across five categories, each age-stratified (5–11, 12–18). An initial 1000-example release was superseded by a quality-curation pass adding a training-only `edge_cases` category (50) and 73 diversified examples, for 1123 total, split 562/561, and 100 new validation examples; Section 4.3 documents how this revision affected earlier results. Examples are instruction/response pairs, without a system prompt, so the model encodes register in its weights.

guard-bear's dataset combines synthetic safe/unsafe examples with unsafe examples from public prompt-injection and jailbreak datasets: 4,089 raw examples, deduplicated to 4,053 unique (2,055 unsafe, 1,998 safe) across 14 subcategories (taxonomy in the repository). "Safe" means *not actively harmful or adversarial*, not "in *small-bear*'s scope"; out-of-scope questions are handled downstream. Dedup precedes splitting, so the 2,836/609/608 train/val/test split (Section 4.3) is disjoint by exact text.

4.3 Base Models, LoRA Configuration, and Training Protocol

We use the Llama 3.2 Instruct family [3] at 1B/3B, 4-bit quantized via `mlx_lm` [14] on Apple Silicon; full-parameter fine-tuning of the 3B model exceeds the 16GB ceiling, making LoRA the only viable strategy. *guard-bear* fine-tunes all 86M parameters via the Metal Performance Shaders backend.

LoRA [4] freezes a pre-trained weight matrix $W_0 \in \mathbb{R}^{d \times k}$ and expresses updates as $\Delta W = BA$, applied to all linear projections in the targeted layers with `scale=20.0` held constant (a raw output multiplier, not an alpha/rank ratio). **Standard** ($r = 8$, `num_layers=16`) covers the upper 16 layers; **Fast** ($r = 4$, `num_layers=8`) covers the upper 8. The two differ in both rank and layer count simultaneously; Sections 3.3–3.4 confirm the resulting crossover is real.

small-bear's adapters train with Adam, cosine-decay LR (peak 1×10^{-5}, 100-step warmup, 600 steps), batch size 4, max sequence length 768, dropout 0.0, loss masked to assistant tokens; all eight runs use seed 42. *guard-bear* fine-tunes `meta-llama/Prompt-Guard-86M` [1] (DeBERTa-based [15]) on the full training split, threshold tuned post-hoc by sweeping $[0.01, 0.95]$ and selecting the best-F1 point meeting ≥ 0.97 recall on the validation split, giving **0.94**.

Reproducibility and dataset vintage. A re-run of the nominal Fast-1B configuration (identical model, data, seed) produced an FK$\leq$7.0 pass rate 17 points lower than an earlier draft's single run. A library-version confound was tested and found insufficient on its own; the training/validation data had also been revised between runs (Section 4.2), and re-scoring the original adapter on the revised validation set alone closed most of the gap. That test is itself imperfect – run against the current, not period-matched, validation set – so it was never cleanly isolated from the vintage confound either. A separate rerun (old-vintage data held fixed throughout, only `mlx` version differing) showed its own genuine, unexplained 16–18-point swing – distinct from the Fast-1B gap above, evidence of real run-to-run instability independent of dataset vintage. The rank sweep, layer sweep, and initial cross-architecture results were found to predate the revision too and were retrained on the current vintage (Table 5's Style-Sep. column excepted, still provisional), motivating the dedicated multi-seed campaign in Section 3.3: a single-run result cannot be trusted without independent scrutiny.

guard-bear received the same scrutiny: splits were audited for exact-text overlap, the dataset deduplicated before splitting to guarantee disjointness by construction, and the model validated with a 5-seed campaign, not a single run (Section 3.2). One wrinkle is worth recording precisely: an initial overlap check compared validation against the full train+val pool it was drawn from, not the actual disjoint training set, making leakage look far more severe (612 of 614) than the true figure (~36 exact-duplicate rows among 4,089 raw examples caused 10 of 614 validation and 3 of 613 test examples to genuinely overlap) – both the measurement error and the real, smaller leak are now fixed at the root.

4.4 Evaluation Framework

For *small-bear*: **readability** via `textstat` [16] (Flesch-Kincaid [17] primary, pass/fail at FK $\leq$ 7.0; SMOG [18], Gunning Fog [19], Coleman-Liau [20], type-token ratio); **latency**; **style separation** via TF-IDF+logistic-regression (scikit-learn [21], 5-fold CV, chance = 0.50). FK $\leq$ 7.0 is a general patient-material health-literacy bar, not a per-age-band fit – looser than the ~grade 5–6 level of the youngest 5–11 patients – used as one shared cutoff across configurations, not a claim of exact age-fit. These are proxies, not direct measures, of age-adapted communication (Section 5). For *guard-bear*: unsafe recall, precision, F1, accuracy, ROC-AUC, and per-subcategory recall.

Ethical approval declarations. Not applicable in the formal IRB/human-subjects sense: no experiments on live vertebrates, humans, or human tissue; all data are synthetic or public reference/prompt-injection materials, with no patient contact or identifying information. This does not certify the deployment context: both systems target a real pediatric population, yet no clinician has reviewed any output (Limitations).

5 Discussion

For *guard-bear*, the central finding is that **a safety classifier cannot be assumed to transfer to a new deployment population without evaluation**; the gap is severe, worse-than-random, not suboptimal thresholding. *small-bear* without an input safety gate is not, by itself, deployable; each component's latency alone is practical on one consumer machine (Section 3.2), though not yet measured combined.

Fast-1B is the best Llama-family fine-tuned configuration for VR deployment (Figure 1): the only fine-tuned option meeting the 1.0s latency target on average with strong readability and style separation – though a single-seed, length-capped prompted baseline (Section 3.1) meets the same readability and latency targets on 1B with no fine-tuning at all, so this recommendation holds only until that comparison is replicated at the multi-seed standard used elsewhere in this paper. SmolLM2 Standard (Section 3.5) beats it on FK pass rate and is a strong alternative where architecture is unconstrained. The **crossover** (Section 3.3) suggests small-model LoRA configuration below 7B does not follow rank-selection guidance developed at larger scales, though only the 1B side is length-independent and its mechanism is only partially resolved (Section 3.4) – necessary but not sufficient for genuine age-adapted communication, since a response can be short and lexically distinct while still emotionally mismatched, a gap only human evaluation can close.

6 Conclusion

This paper shows two things: a general-purpose input-safety classifier does not transfer to a new, vulnerable population without dedicated evaluation (`Prompt-Guard-86M`'s worse-than-random 0.448 AUC, resolved to 0.9992 ± 0.0005 across a 5-seed campaign), and a complete age-adaptive pipeline is achievable on one consumer machine – though a length-capped prompted baseline reaches the same readability and, on 1B, even the same latency, fine-tuning's remaining edge is not depending on a prompt already tuned to the evaluation's thresholds, plus a real 1B style-separation gap (Abstract).

We also uncovered an unexpected LoRA configuration-ordering crossover between 1B and 3B Llama models (Section 3.3), surviving an initial non-reproducible single-seed observation – though only the 1B side is a length-independent register effect, the 3B side being substantially a length effect ($p = 0.055$ once controlled); a rank/layer sweep attributes the crossover partly to rank and depth, though the rank mechanism replicates cleanly only for SmolLM2 – existence established, precise cause only partially so. Neither component alone is deployable: *small-bear* presumes benign input, *guard-bear* makes that presumption safe.

Limitations: Section 4.3 documents this pipeline's non-determinism and dataset-vintage confound, accounted for above; one piece remains provisional (Table 5's Style-Sep. column). The rank sweep's mechanism claim remains qualified at two-seed power (Section 3.4, Conclusion). Length-adjustment was untested on the sweeps' architectures; single-seed perplexity does not track the crossover; models above 3B remain untested. Training and validation data share one generative process, and every metric is fully automated: no human rater has judged any output, so this closed loop confirms internal, not external, validity, and neither component has clinical validation. Most consequentially, no output has ever been checked for factual correctness – readability and style-separation say nothing about whether a response is true. Reported latency excludes *guard-bear*'s pass; its dataset has not been red-teamed, paraphrase-level (as opposed to exact-text) leakage remains unchecked despite the multi-seed campaign, and the provenance check (Section 3.2) has no public-sourced *safe* class to compare against, so it cannot fully rule out a synthetic-vs-public style shortcut specific to the safe class. **Most operationally consequential:** that campaign's per-seed threshold ranges from 0.94 to 0.04 though performance does not – always re-tune on fresh validation data after retraining, which may not always be available. A blocked query returns no response at all (Figure 1); for a distressed child, silence instead of a graceful redirect is itself an unaddressed harm. The prompted-baseline comparison (Section 3.1) tests exactly two prompt variants per role, untuned and one length-capped; this is not a search over prompt space, and the length-capped variant was authored with foreknowledge of the exact FK and latency thresholds being evaluated against, which a deploying engineer would not have without running this same experiment first. Bundling three contributions under one page budget leaves each less room than standalone – guard-bear's related work is one paragraph, and the sweeps run at two seeds versus the response-model campaign's 25–30.

Supplementary information. Validation-set outputs, dataset files, and scripts are available at the repositories under Data availability below; no separate bundle accompanies this submission.

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Declarations

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- **Ethics approval and consent to participate; consent for publication:** Not applicable (no human subjects, patient data, or clinical trial); all data are synthetic or public reference/prompt-injection materials. Not a certification of clinical or deployment readiness – no clinician has reviewed any output (Limitations).
- **Data, materials, and code availability:** *small-bear*'s code/data at https://github.com/abarros6/small-bear; *guard-bear*'s code/data at https://github.com/abarros6/guard-bear.
- **Author contribution:** S.d.R. and R.E. supervised the research. A.B. ran all experiments, designed *guard-bear*, and wrote the paper except the Introduction, written by K.K. All authors reviewed and approved the final manuscript.

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